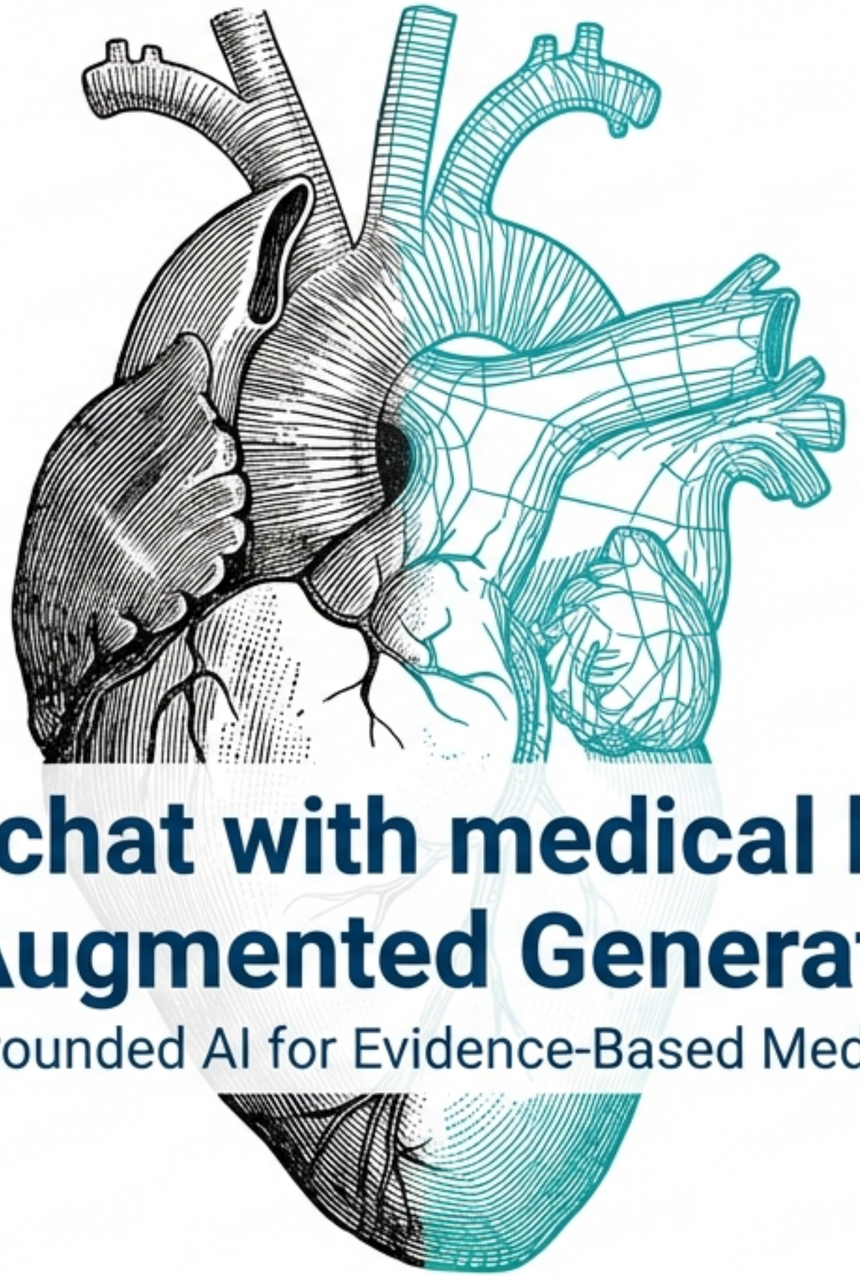


Chatbot to chat with medical books using Retrieval-Augmented Generation Model

MedBuddy: Grounded AI for Evidence-Based Medical Education

NotebookLM



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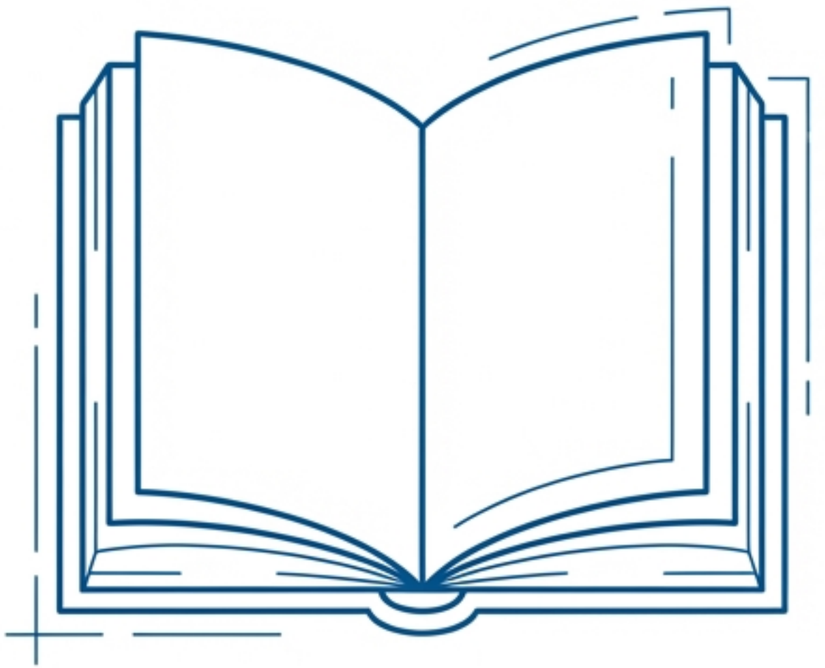
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Institution: Rajadhani Institute of Engineering and Technology (RIET)

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Abstract

Core Technology: This project, MedBuddy, is a digital study assistant designed to help medical students. It uses a technology called Retrieval-Augmented Generation (RAG). In simple terms, RAG is a method where the AI searches for the correct answer in a trusted textbook before responding, rather than just guessing. We built this using IBM Watsonx APIs, which provides enterprise-grade, secure Artificial Intelligence tools, replacing less secure public tools.

Technical Implementation: To build this, we used the Python 3.12 programming language managed by a tool called Miniconda, which ensures the software runs smoothly on any computer. We used a 'Lean AI' strategy, meaning we optimized the heavy software libraries (like PyTorch) to run efficiently on a standard laptop CPU without needing expensive supercomputers.

User Experience: The student interacts with a simple website built with Streamlit. Behind the scenes, we use Industry-Standard Secrets Management to keep the API keys (passwords that let the software talk to IBM's servers) safe and hidden. This ensures the project meets the strict security standards required for handling medical data.

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Introduction

The Project Goal

MedBuddy is developed as part of the CSD 334 Mini Project curriculum. The main goal is to solve the 'Disk Space Trap' and 'Hallucination' problems in AI. 'Hallucination' is when an AI makes up false information. MedBuddy prevents this by using trusted medical books as its only source of truth.

Why IBM Watsonx?

Unlike standard chatbots, we switched to IBM Watsonx. This is an 'Enterprise-Grade' platform, which means it is designed for businesses and professional use. It allows us to use specific AI models (like the Granite-13b-chat) that are optimized to follow instructions carefully and provide concise, accurate answers from the text provided.

The ‘Open Book’ Method

Think of MedBuddy like a student taking an open-book exam.

- **Ingestion**: It reads the textbook (PDF).
- **Indexing**: It organizes the information into a mathematical map (Vector Database).
- **Retrieval**: When asked a question, it finds the exact page.
- **Generation**: It writes an answer based *only* on that page.



MedBuddy

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Introduction

What is LLM (Large Language Model)?



An LLM is a type of Artificial Intelligence that can understand, process, and generate human language. It works like a smart assistant that can read, write, summarize, and answer questions.

The Problem with Standard LLMs



Standard models (like basic ChatGPT) work on probability. If they do not know the answer, they might "Hallucinate," which means they confidently make up false medical information.

What is RAG (Retrieval-Augmented Generation)?



RAG is the 'Open Book' method. Instead of memorizing everything, the AI looks up the specific page in the textbook ***before*** answering. This eliminates hallucinations and ensures the answer is true to the source.

Objectives



- To build a grounded AI chatbot.
- To retrieve accurate clinical information from PDF textbooks.
- To generate context-aware answers using IBM Watsonx.
- To improve learning efficiency for medical students.
- To provide grounded medical answers with citations.

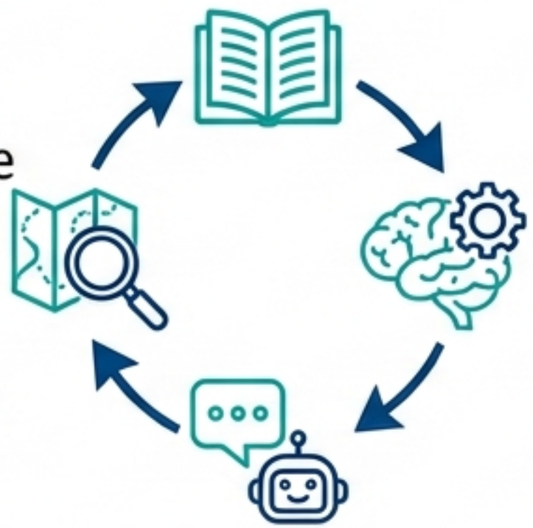
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Core Datasets Used

- **PubMed**: A massive digital library of biomedical literature from the National Institutes of Health. It is used to teach the AI general medical language.



- **Meditrin**: A specialized dataset containing specific medical definitions and clinical instructions, ensuring the AI understands doctors’ terminology.



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Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
6	**The Role of ChatGPT in Medical Education [2023]**	Gunther Eysenbach	Investigated the capabilities of large language models like ChatGPT in clinical settings.	Raised significant concerns regarding content authenticity and the risk of 'hallucinations' (fake facts).
7	**Exploring AI Chatbots in Public Health [2023]**	Francesco Baglivo et al.	A feasibility study on using chatbots for vaccination training.	Requires heavy professional verification to avoid the risk of spreading misinformation.
2	**Ontology Based Personal Assistive Application [2023]**	N. V. Rohil & A. Nair	Used semantic reasoning (understanding meaning) to help elderly people with daily tasks.	Focused on personal daily activity logs rather than academic medical education.

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Literature Survey

Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
3	**Farm-Haystack QA System [2022]**	R. S. Nambiar et al.	Integrated Natural Language Processing (NLP) to answer pandemic-related health questions.	Requires heavy 'fine-tuning' (training) for each specific topic and lacks broad clinical utility.
4	**HR based Chatbot using Deep Neural Network [2022]**	P. C. Nair et al.	Used Deep Learning and the Adam optimizer to help with Human Resources communication.	Designed specifically for HR tasks; lacks the specialized medical knowledge required for healthcare.
5	**AI Chatbot for Nursing Students [2022]**	Jeong-Won Han et al.	Developed a chatbot program to improve nursing skills for electronic fetal monitoring.	Restricted to a single nursing skill focus; cannot answer general medical retrieval questions.

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Literature Survey

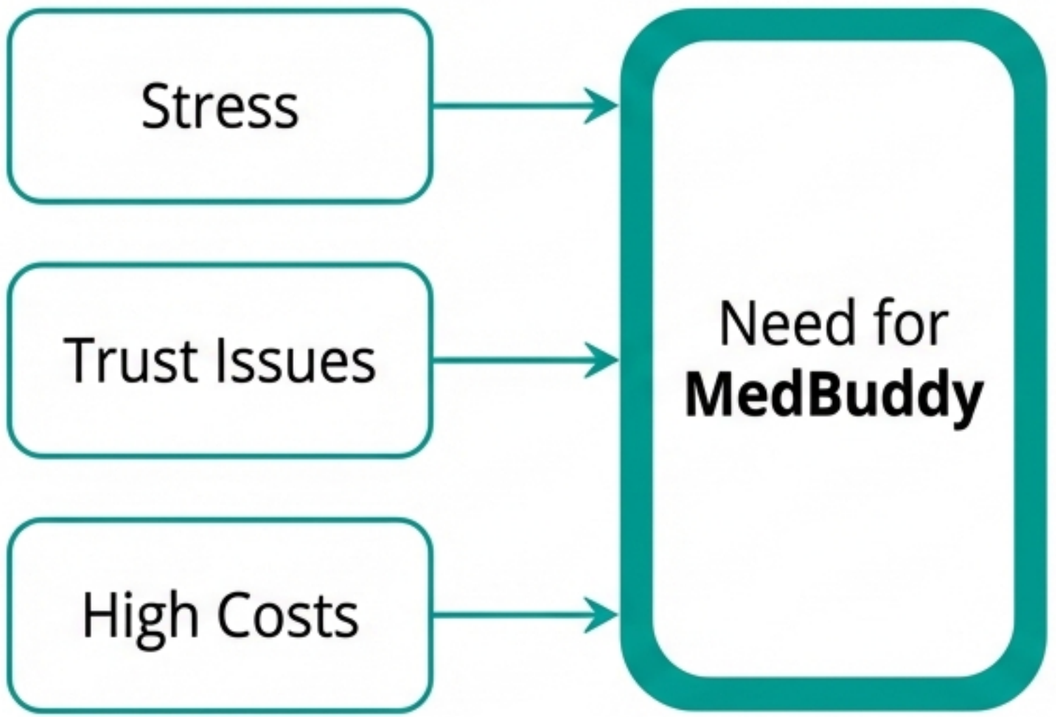
Sl No.	Title [Year]	Author	Proposed Method	Disadvantages
1	**ChEMBL Bot: Chat Bot for ChEMBL Database [2018]**	V. Murali et al.	Created a chat interface to ask questions about chemical compounds and their targets.	Limited only to chemical data queries; it cannot provide guidance from clinical textbooks.

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Problem Identification

- **Information Overload**: Medical students face massive workloads. Searching through 1,000+ page PDFs manually is slow and stressful.
- **The Trust Gap (Hallucinations)**: Generic AI tools often guess when they don't know the answer. In medicine, a 'guess' is a safety risk.
- **Inefficient Search**: Standard 'Ctrl+F' keyword searches fail to understand the context of complex clinical questions (e.g., matching symptoms to a diagnosis).
- **Resource Inequality**: High-quality, updated medical textbooks are expensive. This limits access for students in resource-constrained areas.

The Clinical Knowledge Gap



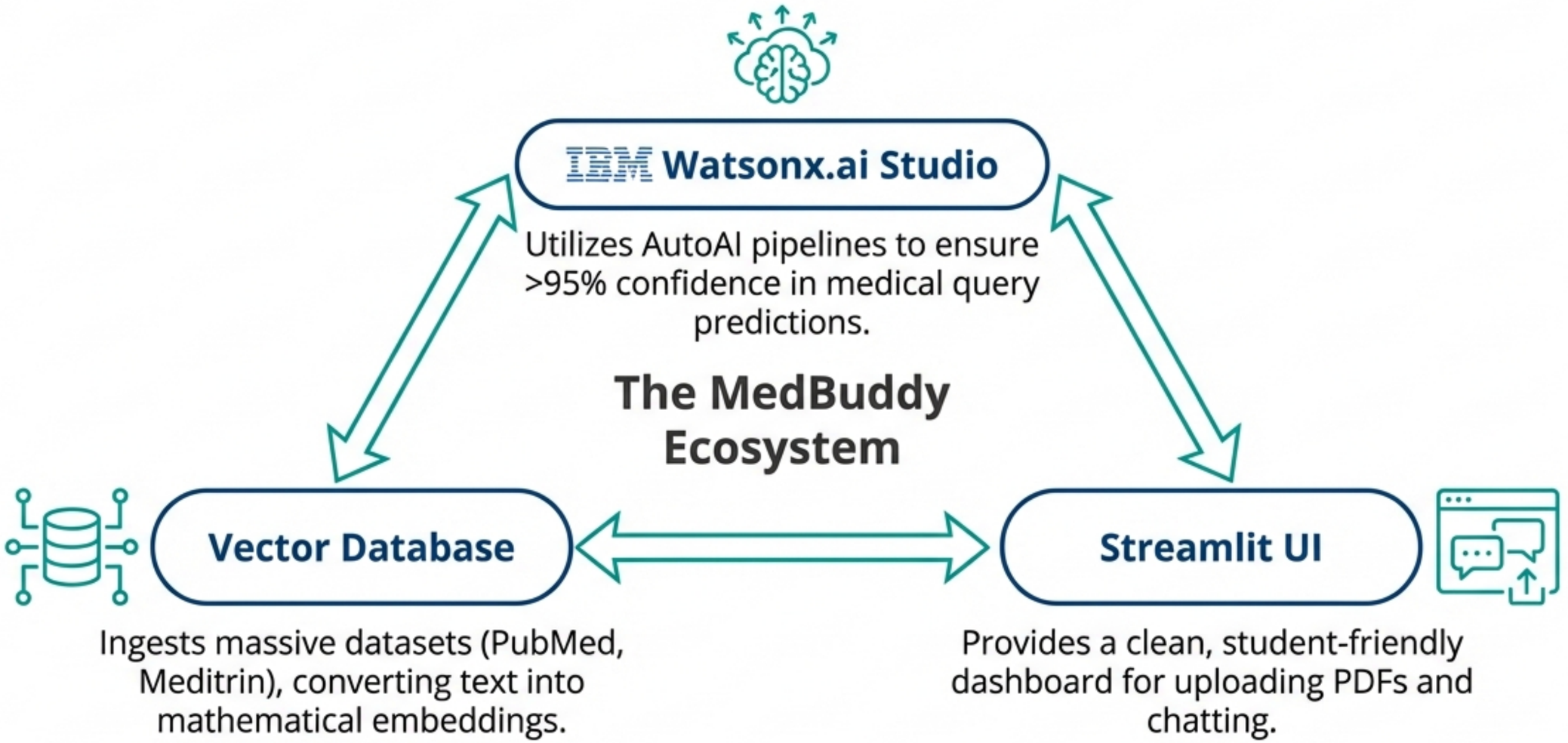
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Proposed System

- **Embeddings**: These are the core of our search. We convert text into long lists of numbers (vectors). Text with similar meanings will have similar numbers, allowing the computer to understand concepts, not just words.
- **Vector Database**: A special storage system that holds these embeddings. It allows us to find the most relevant paragraph in a massive textbook in milliseconds.
- **Semantic Search**: Unlike a keyword search, this finds matches based on *meaning*. For example, searching ‘heart attack’ will also find ‘myocardial infarction’ because the embeddings are close together.
- **Granite-13b-chat**: The specific AI model we use from IBM. It is ‘Instruction Tuned,’ meaning it is trained to follow our command: ‘Answer strictly using the provided context.’

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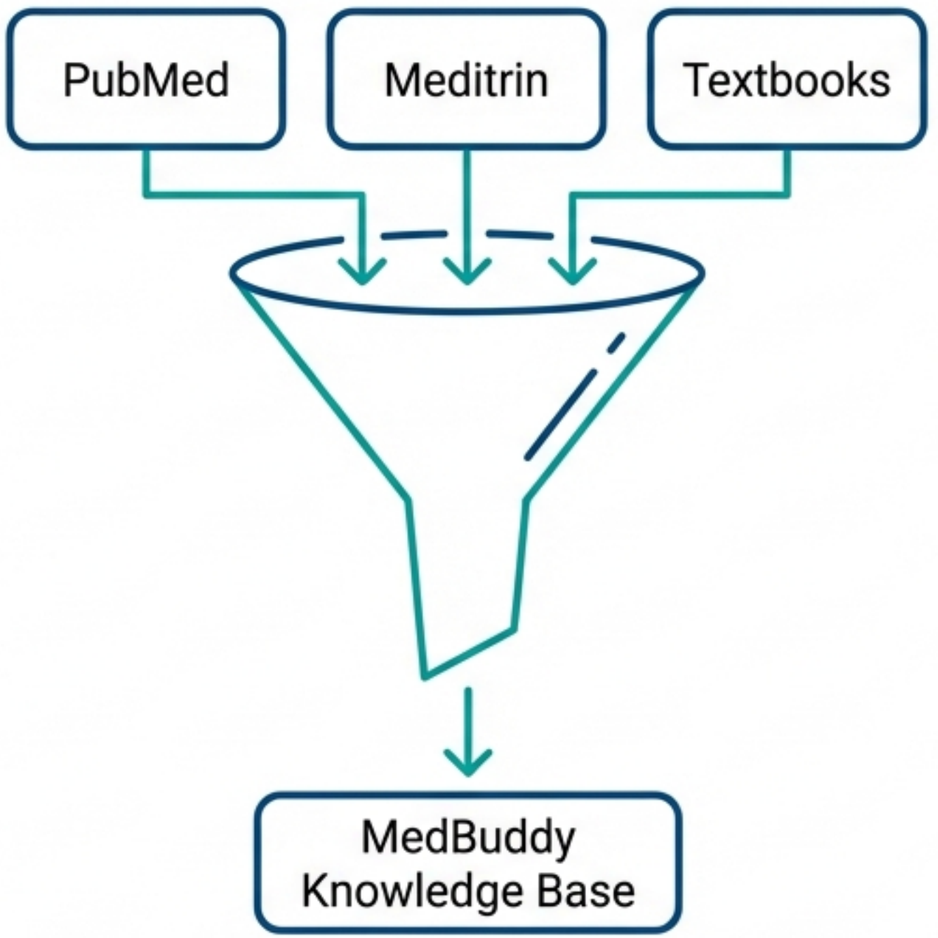
Proposed System



Security: HIPAA-Compliant **Architecture:** Deployed on Watsonx's secure cloud with encrypted data handling.

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Proposed System



Data Integration:

We utilize **PubMed** (for research abstracts) and **Medittrin** (for clinical data) to ground the AI.

Libraries Used:

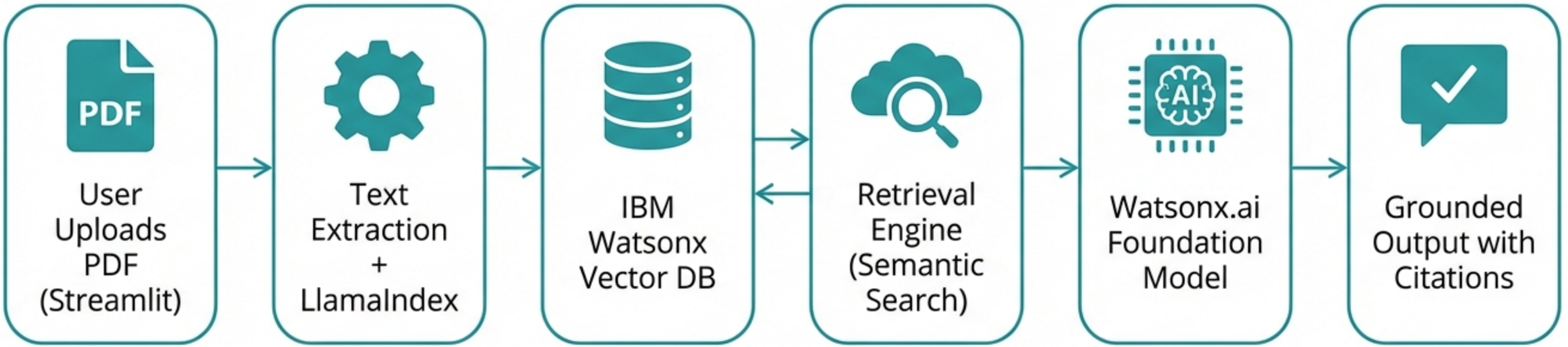
- **LlamaIndex:** A data framework that acts as the bridge between our PDF data and the AI model.
- **ChromaDB / VectorStore:** The technology used to store the mathematical map of the book.
- **HuggingFace Embeddings:** A library that provides the 'MiniLM' model to turn text into numbers efficiently on a laptop CPU.

Security:

By using local secrets management, we ensure that while we use cloud intelligence, our access keys remain private and secure.

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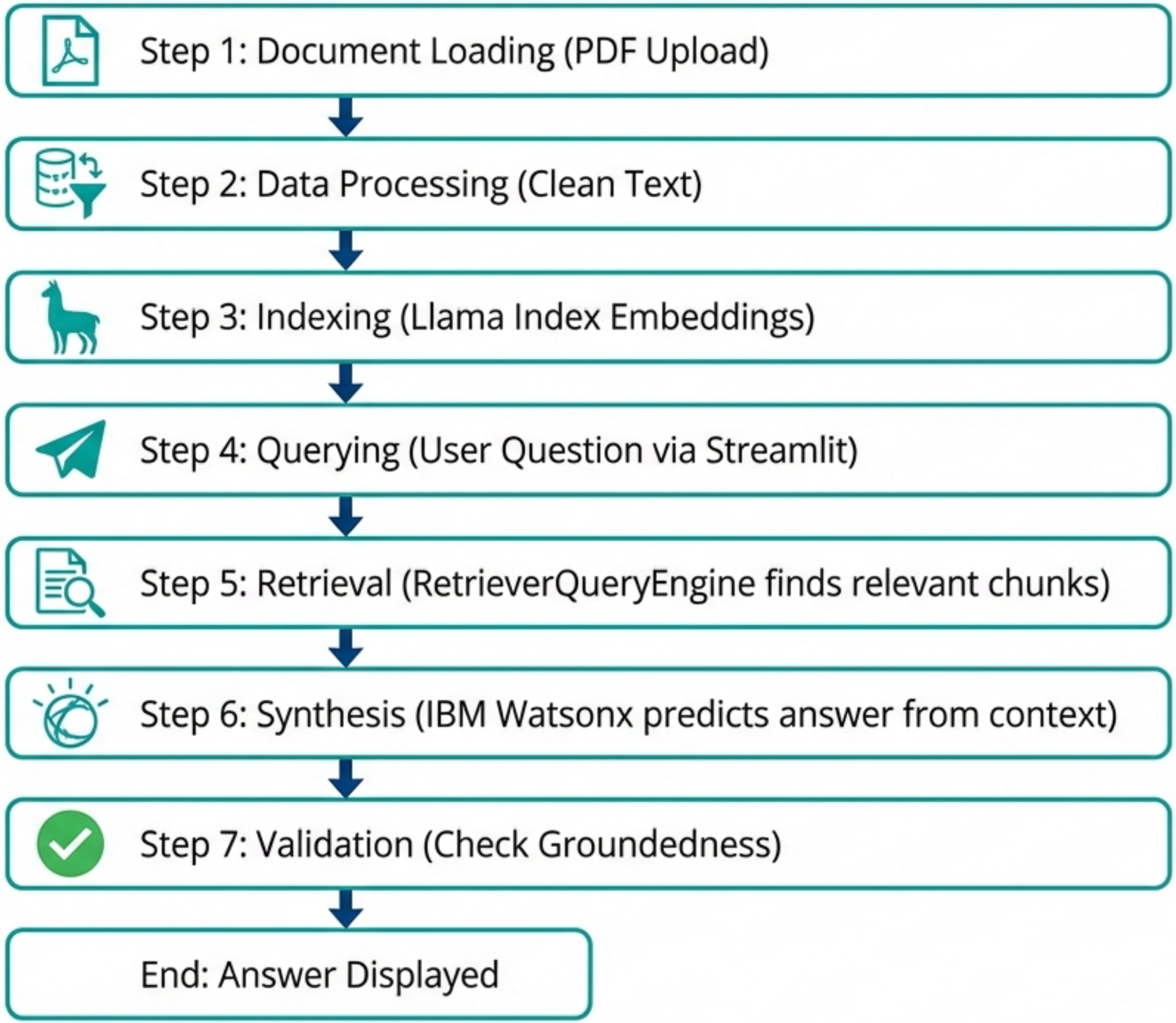
Architecture



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Algorithm

Algorithm

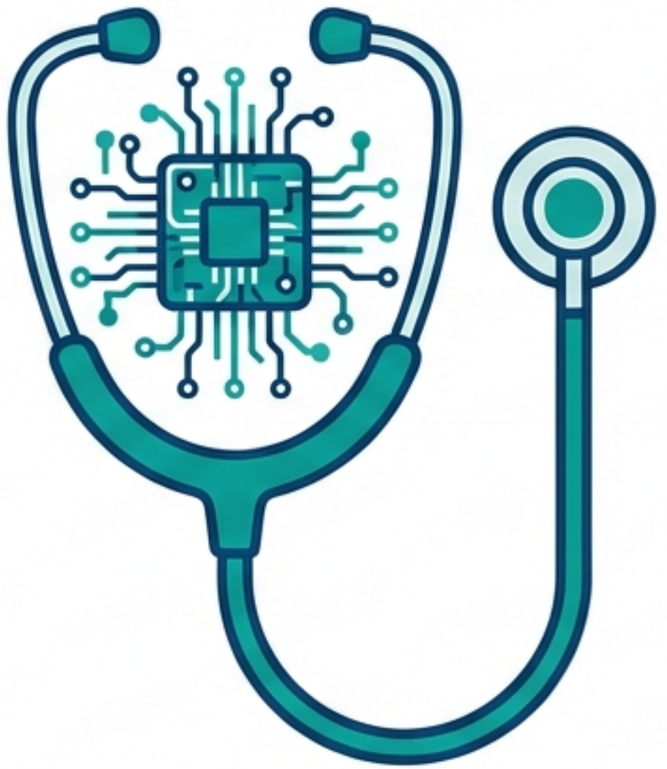


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System and Software Requirements (Hardware)

Laptop Hardware (Minimum):

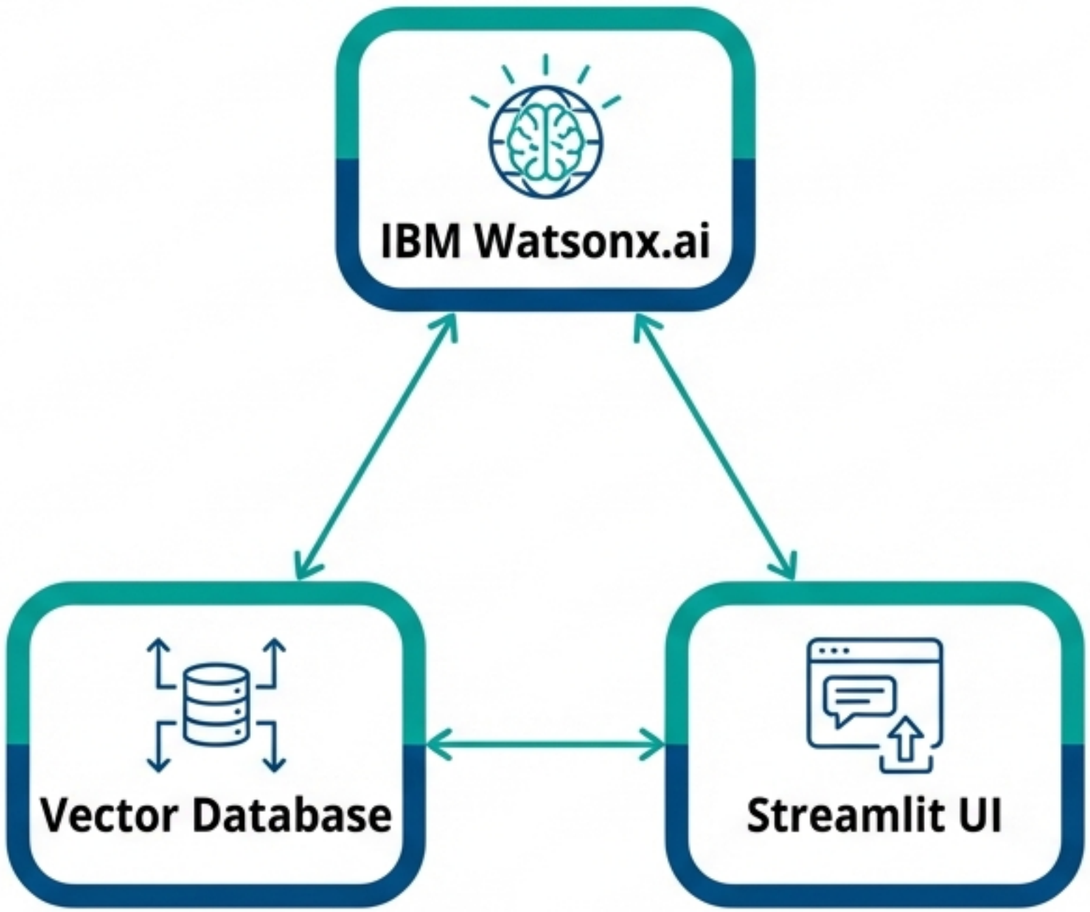
- **Processor:** Intel Core i5 or equivalent (Optimized for CPU usage).
- **RAM:** 8 GB (To handle the Vector Database).
- **Storage:** 256 GB SSD (For storing PDF datasets).
- **Internet:** Stable connection for IBM Watsonx API calls.



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System and Software Requirements

(Software)



- **Front End:** Streamlit (Electric Green). A Python library used to build the user interface. It creates the chat box and upload buttons.
- **Back End:** Python 3.12 (Cyan). The core programming language. Managed by Miniconda (Electric Green) to keep libraries organized.
- **AI Engine:** IBM Watsonx.ai (Cyan). The cloud platform providing the LLM.
- **Vector Store:** ChromaDB (Green). Stores the mathematical map of the book.
- **Libraries:** PyTorch (Green) (for calculations), LlamaIndex (Cyan) (for data connecting).

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Project Plan

Jan 16 - Jan 31, 2026:
Data Ingestion & Indexing

- Implement upload pipeline
- Process PDFs to Vector Database

Feb 01 - Feb 15, 2026:
Retrieval Engine Development

- Develop fetch logic
- Connect to IBM Watsonx

Feb 16 - Feb 28, 2026:
UI Construction & Integration

- Build Streamlit interface
- Backend-Frontend integration

Mar 01 - Mar 15, 2026:
Testing & Final Deployment

- Accuracy testing
- Launch on secure cloud

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Future Plan

Immediate Impact:



70% reduction in search time.



Direct mitigation of academic stress.



Democratization of expensive medical knowledge.

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